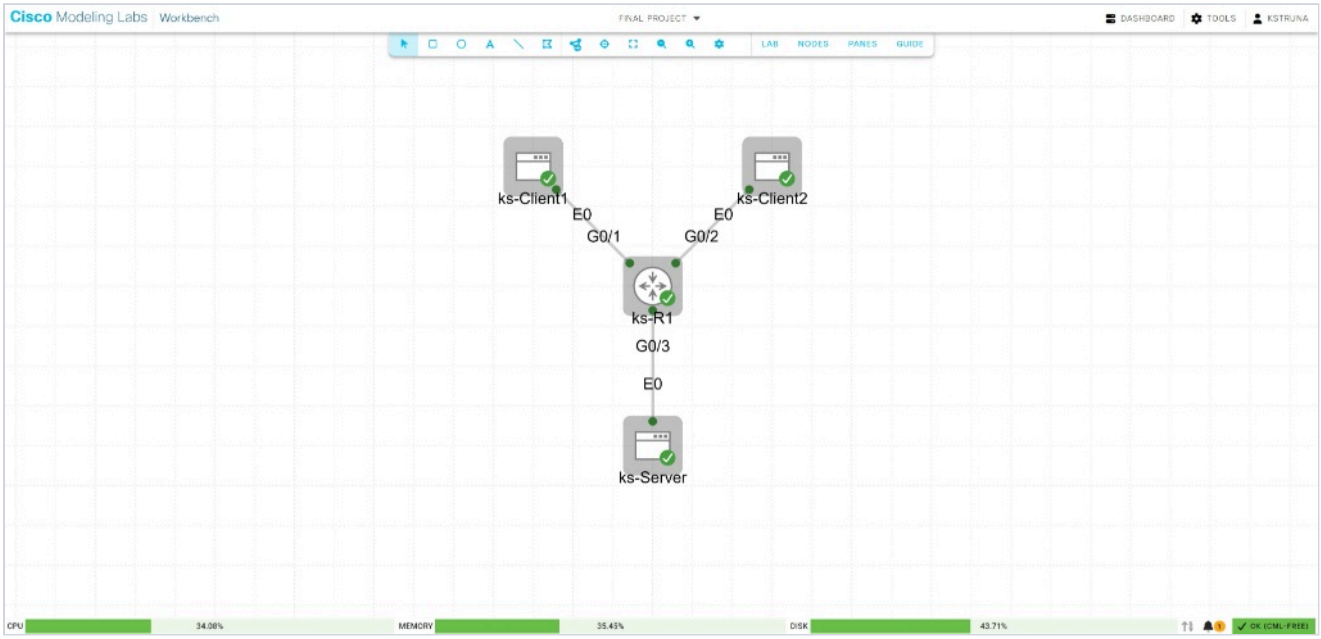
CS 457 - Sprint 4 Network Verification Report

Name: Kevin Struna
Target Domain: `server.struna.edu` (`192.168.20.100`)

1. Running CML Topology

Single-router CML topology with `ks-R1` connected to `ks-Client1` (Gi0/1), `ks-Client2` (Gi0/2), and `ks-Server` (Gi0/3).



2. Router CLI Verification (`ks-R1`)

2.1 `show ip interface brief`

```
ks-R1>show ip interface brief
Interface                  IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0         unassigned      YES unset  administratively down down
GigabitEthernet0/1         192.168.10.1    YES TFTP   up                    up
GigabitEthernet0/2         192.168.11.1    YES TFTP   up                    up
GigabitEthernet0/3         192.168.20.1    YES TFTP   up                    up
```

2.2 `show ip route`

```
ks-R1>show ip route
Gateway of last resort is not set

      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, GigabitEthernet0/1
L        192.168.10.1/32 is directly connected, GigabitEthernet0/1
      192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.11.0/24 is directly connected, GigabitEthernet0/2
L        192.168.11.1/32 is directly connected, GigabitEthernet0/2
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected, GigabitEthernet0/3
L        192.168.20.1/32 is directly connected, GigabitEthernet0/3
```

2.3 `show ip dhcp binding`

```
ks-R1>show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.10.2        0152.5400.cd2b.2f       Nov 11 2026 10:12 PM    Automatic
192.168.11.2        0152.5400.7faf.65       Nov 11 2026 10:12 PM    Automatic
```

3. Game Server Static IP & Default Gateway (`ks-Server`)

```
inserthostname-here:~$ sudo ip link set eth0 up && sudo ip route add default via
192.168.20.1 && ip addr show eth0 && ip route
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen
1000
    link/ether 52:54:00:09:98:88 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.100/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::5054:ff:fe09:9888/64 scope link
       valid_lft forever preferred_lft forever
default via 192.168.20.1 dev eth0
192.168.20.0/24 dev eth0 scope link  src 192.168.20.100
```

4. Client DHCP, DNS Resolution (`nslookup`), and `ping`

4.1 Client 1 (`ks-Client1 - 192.168.10.2`)

```
inserthostname-here:~$ sudo ip link set eth0 up && sudo udhcpc -i eth0 && ip addr
show eth0 && ip route && cat /etc/resolv.conf
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.10.2, server 192.168.10.1
udhcpc: lease of 192.168.10.2 obtained from 192.168.10.1, lease time 86400
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen
1000
    link/ether 52:54:00:cd:2b:2f brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.2/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::5054:ff:fecd:2b2f/64 scope link
       valid_lft forever preferred_lft forever
default via 192.168.10.1 dev eth0  metric 202
192.168.10.0/24 dev eth0 scope link  src 192.168.10.2
nameserver 192.168.10.1

inserthostname-here:~$ nslookup server.struna.edu
Server:         192.168.10.1
Address:        192.168.10.1:53

Non-authoritative answer:
Name:   server.struna.edu
Address: 192.168.20.100

*** Can't find server.struna.edu: No answer

inserthostname-here:~$ ping -c 4 192.168.20.100
PING 192.168.20.100 (192.168.20.100): 56 data bytes
64 bytes from 192.168.20.100: seq=0 ttl=42 time=11.417 ms
64 bytes from 192.168.20.100: seq=1 ttl=42 time=3.215 ms
64 bytes from 192.168.20.100: seq=2 ttl=42 time=3.204 ms
64 bytes from 192.168.20.100: seq=3 ttl=42 time=3.271 ms

--- 192.168.20.100 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 3.204/5.276/11.417 ms
```

4.2 Client 2 (`ks-Client2 - 192.168.11.2`)

```
inserthostname-here:~$ sudo ip link set eth0 up && sudo udhcpc -i eth0 && ip addr
show eth0 && ip route && cat /etc/resolv.conf
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.11.2, server 192.168.11.1
udhcpc: lease of 192.168.11.2 obtained from 192.168.11.1, lease time 86400
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen
1000
    link/ether 52:54:00:7f:af:65 brd ff:ff:ff:ff:ff:ff
    inet 192.168.11.2/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::5054:ff:fe7f:af65/64 scope link
       valid_lft forever preferred_lft forever
default via 192.168.11.1 dev eth0  metric 202
192.168.11.0/24 dev eth0 scope link  src 192.168.11.2
nameserver 192.168.11.1

inserthostname-here:~$ nslookup server.struna.edu
Server:         192.168.11.1
Address:        192.168.11.1:53

Non-authoritative answer:
Name:   server.struna.edu
Address: 192.168.20.100

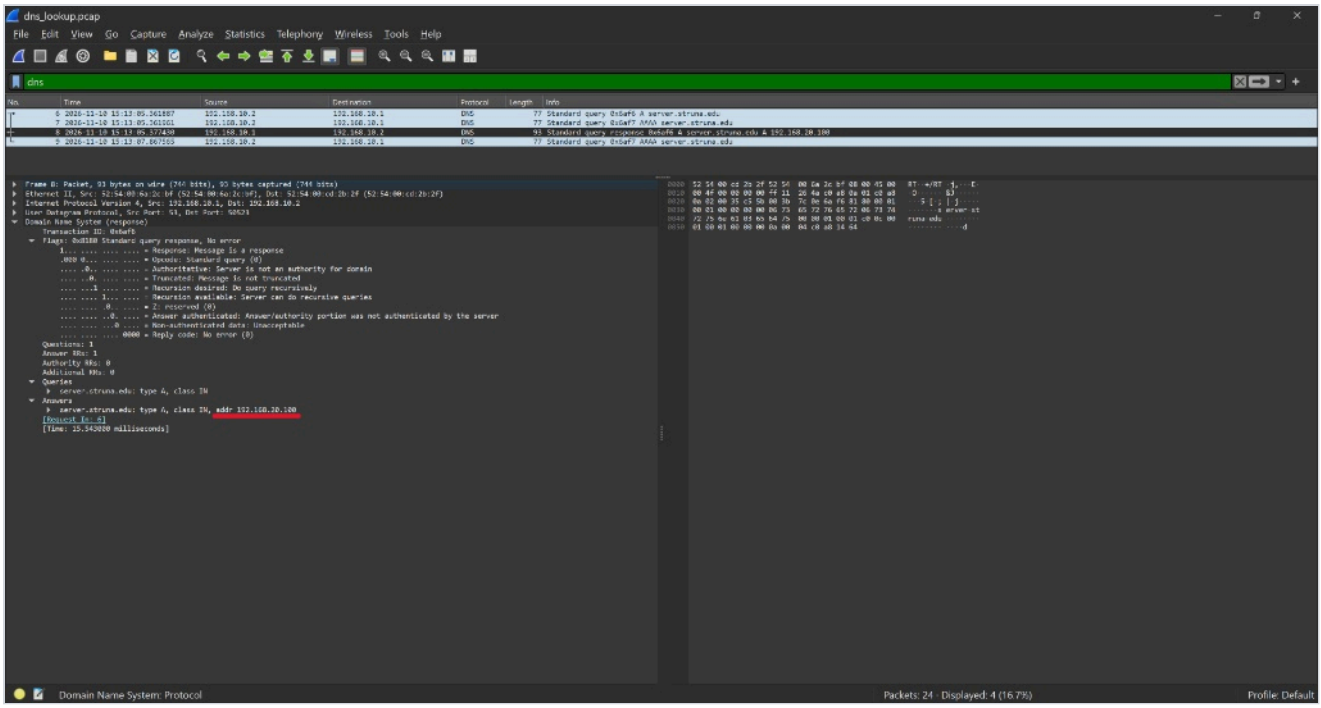
*** Can't find server.struna.edu: No answer

inserthostname-here:~$ ping -c 4 192.168.20.100
PING 192.168.20.100 (192.168.20.100): 56 data bytes
64 bytes from 192.168.20.100: seq=0 ttl=42 time=3.214 ms
64 bytes from 192.168.20.100: seq=1 ttl=42 time=3.410 ms
64 bytes from 192.168.20.100: seq=2 ttl=42 time=3.169 ms
64 bytes from 192.168.20.100: seq=3 ttl=42 time=3.622 ms

--- 192.168.20.100 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 3.169/3.353/3.622 ms
```

5. Wireshark DNS Packet Capture (`dns_lookup.pcap`)

- **Frame 6 (DNS Standard Query):** `192.168.10.2` to `192.168.10.1` (UDP port 53) querying the `A` record for `server.struna.edu`.
- **Frame 8 (DNS Standard Response):** `192.168.10.1` to `192.168.10.2` returning `192.168.20.100`.


dns_lookup.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
6 2026-11-10 15:13:05.561807 192.168.10.2 192.168.10.1 DNS 77 Standard query 0x0af6 A server.struna.edu
7 2026-11-10 15:13:05.561101 192.168.10.1 192.168.10.3 DNS 77 Standard query 0x0af7 AAAA server.struna.edu
8 2026-11-10 15:13:05.572439 192.168.10.3 192.168.10.2 DNS 93 Standard query response 0x0af6 A server.struna.edu A 192.168.20.100
9 2026-11-10 15:13:07.667569 192.168.10.2 192.168.10.1 DNS 77 Standard query 0x0af7 AAAA server.struna.edu
Frame 8: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits)
Ethernet II, Src: 52:54:00:6a:2c:bf (52:54:00:6a:2c:bf), Dst: 52:54:00:cd:2b:2f (52:54:00:cd:2b:2f)
Internet Protocol Version 4, Src: 192.168.10.1, Dst: 192.168.10.2
User Datagram Protocol, Src Port: 53, Dst Port: 50521
Domain Name System (response)
Transaction ID: 0x0af6
Flags: 0x8180 Standard query response, No error
1... .... = Response: Message is a response
.000 0... .... = Opcode: Standard query (0)
.... .0.. .... = Authoritative: Server is not an authority for domain
.... ..0. .... = Truncated: Message is not truncated
....1 .... = Recursion desired: Do query recursively
.... 1... .... = Recursion available: Server can do recursive queries
.... .0.. .... = Z: reserved (0)
....0. .... = Answer authenticated: Answer/authority portion was not authenticated by the server
....0 .... = Non-authenticated data: Unacceptable
.... 0000 = Reply code: No error (0)
Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 0
Queries
server.struna.edu: type A, class IN
Answers
server.struna.edu: type A, class IN, addr 192.168.20.100
[Request In: 6]
[Time: 15.543020 milliseconds]
Domain Name System: Protocol
Packets: 24 · Displayed: 4 (16.7%)
Profile: Default